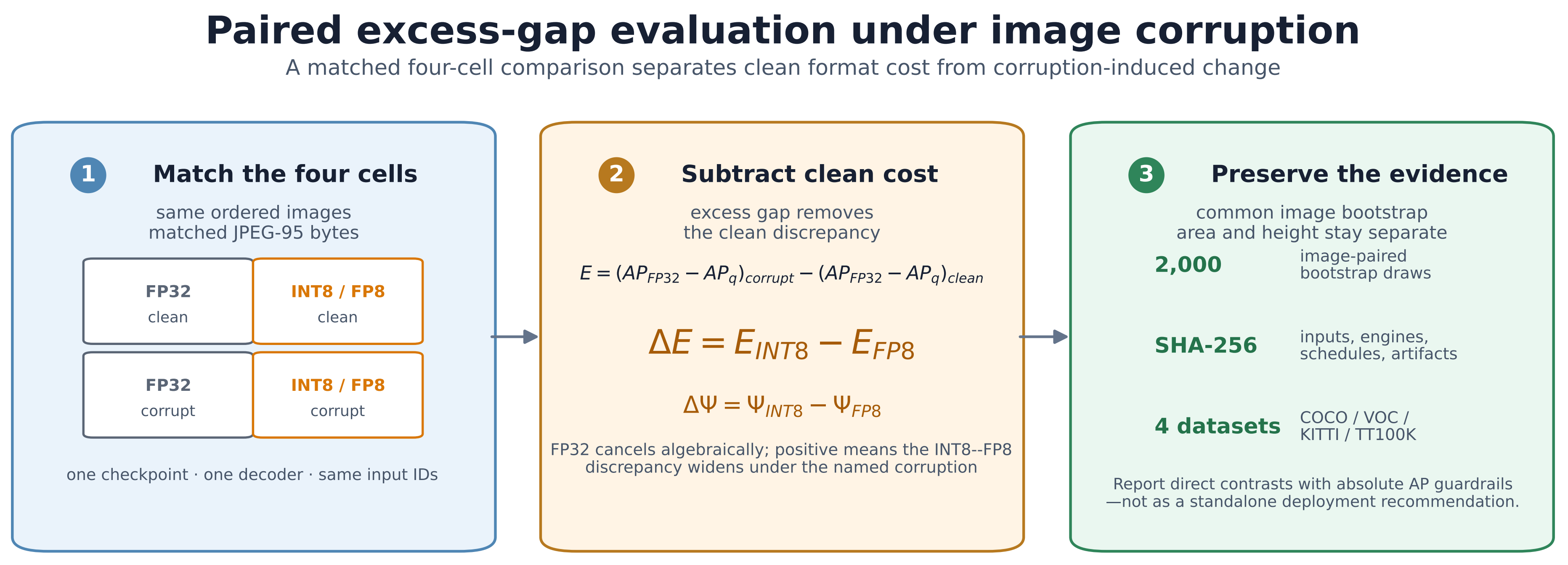
Paired excess-gap evaluation under image corruption
A matched four-cell comparison separates clean format cost from corruption-induced change
1
Match the four cells
same ordered images
matched JPEG-95 bytes
FP32
clean
INT8 / FP8
clean
FP32
corrupt
INT8 / FP8
corrupt
one checkpoint · one decoder · same input IDs
2
Subtract clean cost
excess gap removes
the clean discrepancy
$E = (AP_{FP32} - AP_q)_{corrupt} - (AP_{FP32} - AP_q)_{clean}$
$\Delta E = E_{INT8} - E_{FP8}$
$\Delta\Psi = \Psi_{INT8} - \Psi_{FP8}$
FP32 cancels algebraically; positive means the INT8--FP8
discrepancy widens under the named corruption
3
Preserve the evidence
common image bootstrap
area and height stay separate
2,000
image-paired
bootstrap draws
SHA-256
inputs, engines,
schedules, artifacts
4 datasets
COCO / VOC /
KITTI / TT100K
Report direct contrasts with absolute AP guardrails
—not as a standalone deployment recommendation.